

Response to the Referee Reports

Manuscript: “Noise-Assisted Feedback Control of Open Quantum Systems for Ground State Properties”
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Dear Referees,

We sincerely thank both the referees for carefully reading our manuscript and for providing constructive feedback. Please find below a detailed point-by-point response. For each item, we provide (i) your comment, (ii) our response, and (iii) the changes made in the revised manuscript.

RESPONSE TO REFEREE 1

The manuscript introduces a noise-assisted feedback-based quantum algorithm (NAFQA) designed to steer an open quantum system toward a virtual ground state. The approach combines quasiprobability decomposition (QPD) to implement effective non-Markovian dynamics with Lyapunov-based control to enforce monotonic energy decrease. The protocol is demonstrated on small instances of Max-Cut and spin-glass Hamiltonians. The topic is timely and relevant to near-term quantum computing, particularly in the context of dissipative algorithms. The manuscript is generally well written and technically careful. The idea is conceptually interesting, I therefore recommend acceptance given the comments below are addressed appropriately.

Response. We thank the referee for finding our work conceptually interesting and for their positive assessment. We address their comments below and hope that they will find our responses satisfactory and recommend the revised manuscript for publication.

Major: “1. Examples Considered — Since the results are largely based on numerical simulations, it would be helpful for the authors to provide additional details about the instances used. For the Max-Cut problem, it is important to specify the connectivity of the graph and whether it is frustrated, as numerical performance can depend strongly on the structure of the instance. Similarly, the spin-glass example uses only three qubits, which may limit the insight one can draw. It would be useful if the authors could clarify the choice of such small system sizes and discuss the implications for the generality of the results.”

Response: We thank the referee for raising this point. In the case of *Max-Cut problem*, we have used the graph with a 5-qubit linear chain with edges $\{(1, 2), (2, 3), (3, 4), (4, 5)\}$, motivated by the original FQA paper by Magann et al. [1]. This graph is non-frustrated, as it contains no odd cycles. We note that the noise model is independently taken from the 5-qubit IBM-Vigo emulator, as shown in Fig. 6(c); therefore, the graph topology and the device noise model are distinct choices. We have added this specification to the revised manuscript.

In the case of *Spin-glass model*, previously, we demonstrated our simulations for $N = 3$ qubits. In response to the referee comment, we have expanded our simulations to $N = 5$ qubits with all-to-all connectivity, averaging over 25 independent disorder realizations (seeds 0 through 24) with couplings J_{mn} and local fields h_l drawn uniformly from $[-1, +1]$. The NAFQA protocol continues to demonstrate faster convergence to the approximate virtual ground state compared to noisy FQA at $N = 5$, consistent with the $N = 3$ results. Additionally, we have included NAFQA simulations for $N = 8$ qubits using a synthetic sparse Pauli-Lindblad noise model (not with a device-characterized noise model), with results reported in the Supplementary Information (SI). These results further corroborate the performance and robustness of NAFQA across system sizes.

Finally, to ensure full reproducibility of all numerical results, the code and data required to reproduce every figure in the manuscript are publicly available on the GitHub repository [2].

Changes Made: In the section “Optimization problem,” we have now clarified the connectivity and the graph topology of the maxcut problem. Further, we have provided additional simulations for the spin-glass model for $N = 5$ qubits with all-to-all connectivity, averaged over 25 disorder realizations (seeds 0 through 24), with results presented in Fig. 5. Additional results for $N = 8$ qubits using a synthetic sparse Pauli-Lindblad noise model are provided in the SI. Moreover, the code and data required to reproduce every figure in the manuscript are publicly available on the GitHub repository [2].

Minor: “Novelty and comparison to dissipative algorithms – The manuscript combines two established techniques–Lyapunov-based control and the implementation of non-Markovian dynamics via QPD. Its novelty lies in integrating these components into a ground-state preparation framework. The manuscript could be strengthened by more clearly articulating the conceptual distinction between the proposed method and other dissipative algorithms, for instance dissipative cooling or standard VQE combined with probabilistic error cancellation (PEC).”

Response. We thank the referee for this suggestion. We have followed the advice of the referee and have added a paragraph in the Introduction and an expanded comparison in the Discussion section. The key conceptual distinctions are as follows:

(i) *Distinction from dissipative cooling.* In the standard dissipative cooling or bath-engineering protocols [3–5], the Lindblad operators and decay rates are fixed and designed to have the target state as the unique steady state of the engineered Lindbladian. The protocol is therefore open-loop: the dissipation is not adjusted as a function of the evolving quantum state. In contrast, NAFQA employs a closed-loop feedback-adaptive strategy where at each Trotter layer s , the decay rates $\Gamma_k(s+1)$ are updated according to the measured expectation values via the Lyapunov condition $\Gamma_k(s+1) = -(\langle P_k^\dagger H_p P_k \rangle - \langle H_p \rangle)_s$. This layer-dependent, state-dependent tuning is the defining feature of NAFQA and is absent in passive dissipative cooling. Furthermore, NAFQA does not require prior knowledge of the ground state to design the dissipation; the feedback law is constructive and derived entirely from the Hamiltonians H_p , H_d , and the available noise operators P_k .

(ii) *Distinction from VQE + PEC.* In the variational quantum eigensolver (VQE) augmented with probabilistic error cancellation (PEC), PEC is applied uniformly to suppress the effects of device noise on the measured expectation values, mimicking noiseless simulation at an additional overhead cost. In NAFQA, PEC (specifically, QPD) is used not to suppress noise but to engineer additional non-Markovian dynamics with negative decay rates that are absent from the physical device. The device noise is retained and incorporated into the feedback law rather than cancelled. This constitutes a fundamentally different use of the QPD framework: in VQE+PEC, QPD restores unitary evolution; in NAFQA, QPD augments the physical noise channel to realize engineered open-system dynamics that are inaccessible to purely unitary circuits.

Changes Made: Taking into account all these points, we have added a dedicated comparison paragraph at the opening of the Discussion section.

“Scalability and Exponential Overhead — The manuscript appropriately acknowledges the exponential quantum overhead associated with implementing negative decay rates via QPD. It would be helpful to clarify whether the overhead discussed on pp. 5–6 refers solely to implementing the engineered non-Markovianity, and whether this cost is in addition to the intrinsic overhead of PEC itself. Moreover, it would be useful to explicitly state that there is also classical exponential overhead associated with storing and tracking the sign variable $s(t)$. Given these costs, it is not immediately clear whether introducing non-Markovianity via QPD provides a net practical advantage. A brief discussion clarifying this point would improve the presentation.”

Response: We thank the referee for this careful comment, which allows us to sharpen the presentation of the revised manuscript.

(i) The overhead discussed on pp. 5–6 refers solely to the QPD implementation of the engineered noise map Λ'_s with negative rates $\{\nu_k(t)\}_{k \in \mathcal{K}}$. It is not the standard PEC overhead for mitigating the intrinsic device noise $\{\lambda_k\}$, and the two should not be conflated. In NAFQA, the intrinsic noise channel Λ_s (with positive rates $\{\lambda_k\}$) is deliberately retained and incorporated into the Lyapunov feedback law. It is never canceled or mitigated. Consequently, no PEC overhead associated with $\{\lambda_k\}$ is incurred. The only QPD overhead is that associated with the engineered map Λ'_s , which is governed exclusively by the magnitudes $|\nu_k(t)|$ of the negative rates, as expressed in the corrected Eq. (16) of the revised manuscript.

(ii) The classical cost of the sign variable $s(t) \in \{+1, -1\}$ depends on which of the two implementation pathways of NAFQA is used, and we thank the referee for prompting us to clarify this distinction.

In the QPD method, the sign $s(t)$ is tracked as part of the standard quasi-probability post-processing, the same way as in the PEC literature [6]. For M circuit instances over s layers with $|\mathcal{K}|$ channels, the sign bookkeeping cost is $\mathcal{O}(Ms|\mathcal{K}|)$ classical operations, which is polynomial in all parameters and is subsumed

into the sampling overhead $\mathcal{N}^2$. No additional exponential classical cost arises beyond what is already captured by Eq. (16) of our paper.

In the PLQT pathway, the sign dynamics is governed by the averaged sign [7],

$$\bar{s}(T) = \exp \left[-2 \int_0^T R(\tau) d\tau \right], \quad R(t) = \sum_{k \in \mathcal{K}, \Gamma_k < 0} r_k(t), \quad (1)$$

where $R(t)$ is the total rate of sign-flipping quantum jumps. As $\bar{s}(T) \rightarrow 0$, positive and negative trajectories become balanced and the statistical signal vanishes, requiring $M \propto 1/\bar{s}(T)^2$ trajectories to maintain fixed precision. However, this is not an additional cost beyond the QPD overhead: for the standard choice of jump rates $r_k(t) = |\nu_k|$ and purely negative engineered rates $\nu_k(t) \leq 0$, the sign-flip rate satisfies $R(t) = \sum_{k \in \mathcal{K}} |\nu_k(t)|$, so that

$$\bar{s}(T) = \exp \left[-2 \sum_{k \in \mathcal{K}} \int_0^T |\nu_k(t)| dt \right] = \mathcal{N}(T)^{-1}. \quad (2)$$

The averaged sign in PLQT is therefore exactly the reciprocal of the QPD sampling overhead of Eq. (16). Both implementation methods pay the same fundamental cost $\mathcal{N}^2$. The threshold $|\nu_k(t)| \leq \nu_{\max}$ imposed in our simulations controls $R(t)$ and hence $\bar{s}(T)$ directly, ensuring that both the QPD sampling overhead and the PLQT sign decay remain tractable over the circuit depths considered. In this sense, the exponential cost of tracking $s(t)$ is fully captured by Eq. (16) in both methods, and does not constitute an independent additional overhead.

Importantly, this exponential cost is not an artifact of our particular implementation but is fundamental in nature. The correspondence $\bar{s}(T) = \mathcal{N}(T)^{-1}$ reveals a direct structural analogy with the fermionic sign problem in quantum Monte Carlo simulations [8], where the average sign decays exponentially as $\langle s \rangle \sim e^{-\beta N \Delta f}$ and the required number of samples scales as $1/\langle s \rangle^2$. In both cases, the exponential cost originates from the presence of negative weights in a stochastic decomposition. Since Troyer and Wiese established that the sign problem is NP-hard [8], this analogy suggests that the exponential overhead for simulating negative Lindblad rates is likewise a fundamental barrier rather than a practical limitation of current methods. We emphasize, however, that this analogy is structural and not a formal complexity-theoretic reduction; establishing a rigorous hardness result for the negative-rate Lindblad setting would require a separate proof. The threshold $|\nu_k(t)| \leq \nu_{\max}$ plays the role of a depth cutoff that keeps the simulation within the tractable regime.

(iii) For completeness, the total sampling cost of NAFQA relative to the original FQA cost $N_s^{\text{FQA}} = \mathcal{O}(mds)$ of Refs. [1, 9] is:

$$N_s^{\text{NAFQA}} = \mathcal{O}(mds \cdot |\mathcal{K}| \cdot \mathcal{N}^2(s)), \quad (3)$$

where m is the number of samples per Pauli expectation value, d is the maximum graph degree, and $|\mathcal{K}| \ll 4^N - 1$ under the assumption of the Pauli-Lindblad model. The factor $|\mathcal{K}|$ enters linearly, while $\mathcal{N}^2$ is exponential in $|\nu_k|s$ but is regulated by the threshold $\nu_{\max}$ in our revised manuscript.

The net practical advantage of NAFQA is not a reduction in the sampling cost relative to FQA; it does not reduce it. Rather, NAFQA restores the monotone descent guarantee that is otherwise irreversibly lost when FQA is executed on NISQ hardware. Proposition 1 formally establishes that noisy FQA loses the Lyapunov condition $\frac{d}{dt} \langle H_p \rangle \leq 0$ once $A(t)^2 < \Lambda_{\text{noise}}$, causing the system to drift toward the mixed state (further away from the true ground state) regardless of the sampling budget or circuit depth, precisely the non-monotonic behavior shown in Figs. 2(e) and 5. Theorem 1 establishes that NAFQA recovers an unconditional, hardware-independent monotone descent at all circuit depths and system sizes, independently of $\{\lambda_k\}$, N , and s . This correctness guarantee, which is structurally unavailable to noisy FQA, is the fundamental motivation for our work. The exponential QPD cost is the price paid for algorithmic correctness, in the same spirit as the overhead accepted in standard PEC (error mitigation) for unbiased expectation value estimation [6, 10, 11], but serving a qualitatively different purpose: monotone descent assurance rather than noise cancellation.

Changes made: (i) We have corrected Eq. (16) in the revised manuscript (the original contained a typographical sign error) and added a paragraph below Eq. (16) clarifying the discussions regarding the overhead in implementing the negative rates.

- (ii) We have added a paragraph after Eq. (29) in the Methods section discussing the sign dynamics of the PLQT approach and its similarity to the QPD overhead.
- (iii) We have added a paragraph to the Discussion section that explicitly addresses the advantages and limitations of our algorithm NAFQA.

“Virtual ground state — While the Lyapunov control ensures a local monotonic decrease of the energy, it does not guarantee convergence to the global ground state, particularly in many-body systems with multiple local minima. Explicitly noting this limitation would improve conceptual clarity.”

Response. We thank the referee for this important point, and agree that the convergence structure of NAFQA should be stated with care. In the revised manuscript, we have clarified this as follows.

Theorem 1 establishes a monotone descent,

$$\frac{d}{dt}\langle H_p \rangle = -A(t)^2 - \sum_{k \in \mathcal{K}} C_k(t)^2 \leq 0, \quad \forall t \geq 0, \quad (4)$$

regardless of the Pauli noise coefficients $\{\lambda_k\}$, the system size N , or the circuit depth s . This guarantees monotone energy descent but does not, by itself, guarantee convergence to the global ground state, as pointed out by the referee correctly.

By LaSalle’s invariance principle, the trajectory converges to the largest invariant subset of

$$\mathcal{S} = \{\rho : A(t) = 0 \text{ and } C_k(t) = 0 \forall k \in \mathcal{K}\}. \quad (5)$$

Two conditions, each acting on a different part of the dynamics, restrict $\mathcal{S}$:

- (C1) $\langle i[H_d, H_p] \rangle = 0$ only at energy eigenstates of H_p , which holds when $\{H_p, H_d\}$ generate $su(2^N)$. This restricts $\mathcal{S}$ via the unitary part of the dynamics, analogously to conditions (1)–(3) of Magann et al. [9].
- (C2) For every excited eigenstate $|E_n\rangle$ ($n > 0$), $\exists k \in \mathcal{K}$ such that $\langle E_n | P_k^\dagger H_p P_k | E_n \rangle \neq E_n$, so that the dissipative term removes excited eigenstates from $\mathcal{S}$ [12].

Both conditions are checkable a priori from the operator structure alone and are verified for all MaxCut and spin-glass instances studied here.

We emphasize that a rigorous proof that (C1) and (C2) guarantee $\mathcal{S} = \{|\psi_{\text{gs}}\rangle\langle\psi_{\text{gs}}|\}$ for general mixed-state dynamics remains an open question, as the NAFQA trajectory is generically mixed due to the dissipators. This limitation is now explicitly acknowledged in both the main text and the SI, where we note that a complete characterization of $\mathcal{S}$ for open-system dynamics is left for future work. The referee’s observation has thus directly improved the conceptual clarity of the manuscript.

Changes made: We have added Theorem 1 and the subsequent discussions, including conditions (C1) and (C2), to the revised manuscript and to the SI.

“Typo— Do the authors mean U_s instead of $\tilde{U}_s$ in the paragraph below Eq 2, since the device noise is accounted for in Λ_s ? ”

Response. We thank the referee for pointing out our mistake.

Changes made: We have corrected this error in the revised manuscript.

“5. Introduction— The statement in the introduction that noise is used “as a resource to simulate non-Markovian dynamics” could be phrased more precisely. It is not clear if the device noise in some way helps with implementing the engineered non-Markovianity rather, it seems, it is implemented purely classically using quasiprobability decomposition.”

Response. We thank the referee for raising a legitimate concern about the precision of language. In the introduction, we have now explicitly stated what we mean by the non-Markovian dynamics and mentioned that NAFQA employs a feedback-adaptive strategy that operates directly on the native noise of the device.

Changes made: In the revised Introduction, we have added the following clarifying passage: “By non-Markovian, we specifically refer to dynamics governed by a time-local Gorini–Kossakowski–Sudarshan–Lindblad

(GKSL) master equation with negative decay rates [13], a regime that, while formally unphysical as a standalone noise channel, can be implemented via classical post-processing through the pseudo-Lindblad quantum trajectory (PLQT) [7] or the quasiprobability decomposition (QPD) methods [14, 15]. Importantly, such negative rates are not merely a theoretical construction; recent independent works demonstrate that they are generic in noisy intermediate-scale quantum (NISQ) devices and are necessary for correct noise characterization [16, 17].”

In addition, in the paragraph introducing NAFQA, we have added the following sentence. “Unlike passive dissipative cooling protocols, in which Lindblad operators and decay rates are fixed a priori to engineer a target steady state [3–5], NAFQA employs a closed-loop, feedback-adaptive strategy that operates directly on the native noise of the device; a detailed comparison with dissipative and variational approaches is presented in the Discussion.”

RESPONSE TO REFEREE 2

“The authors propose a hybrid approach combining feedback-based quantum algorithms with engineered dissipation, including negative rates, to prepare ground states on noisy devices. The idea of exploiting intrinsic noise as a resource, rather than suppressing it, is conceptually innovative. However, the manuscript has two critical weaknesses that limit its impact:

Response. We thank the referee for the detailed and thoughtful report and for the positive assessment of the technical soundness, novelty, and relevance of the manuscript. We welcome the opportunity to address their two critical concerns and specify the practical advantage of our algorithm. We believe that the revised manuscript substantially strengthens both the theoretical foundations and the contextual positioning of the work.

(i) Lack of rigorous convergence guarantees to the true ground state. Numerical results (Fig. 2c) show the success probability saturates well below 1, indicating convergence to a virtual steady state, not the true ground state. In closed systems, Lyapunov control only guarantees convergence to an invariant set (LaSalle), which includes excited states. In the open-system setting, the final state is determined by the interplay of unitary and dissipative dynamics, with no guarantee that the attractor coincides with the global minimum. The authors provide no systematic conditions under which the true ground state is reached, leaving the algorithm as a heuristic whose success is problem-dependent.”

Response. We thank the referee for this criticism, which directly motivated the most significant theoretical addition in the revised manuscript. In response, we have introduced Proposition 1 and Theorem 1, which together sharpen the theoretical foundations of NAFQA and address the referee’s concern directly.

(i) **Monotone descent (Theorem 1).** Under the NAFQA joint feedback law $\beta(t) = -A(t)$ and $\Gamma_k(t) = -C_k(t)$, the energy derivative satisfies

$$\frac{d}{dt}\langle H_p \rangle = -A(t)^2 - \sum_{k \in \mathcal{K}} C_k(t)^2 \leq 0, \quad \forall t \geq 0, \quad (6)$$

at every Trotter layer, regardless of system size, circuit depth, or hardware error rates $\{\lambda_k\}$. This is qualitatively unavailable to noisy FQA: Proposition 1 establishes that the energy in noisy FQA decreases monotonically only when $A(t)^2 \geq \Lambda_{\text{noise}}$, where $\Lambda_{\text{noise}} = 2\|H_p\| \sum_{k \in \mathcal{K}} \lambda_k$ is a fixed hardware-determined constant. Once $A(t)^2 < \Lambda_{\text{noise}}$, the noise term dominates, and the system drifts irreversibly toward a mixed state that is further away from the ground state — precisely the non-monotonic behavior visible in Figs. 2(e), 4, and 5 of the revised manuscript.

However, monotone descent alone does not preclude convergence to excited states, as the referee correctly observes. To address this, we apply LaSalle’s invariance principle: NAFQA converges to the largest invariant subset of

$$\mathcal{S} = \{\rho : A(t) = 0 \text{ and } C_k(t) = 0 \forall k \in \mathcal{K}\}. \quad (7)$$

Two conditions, each acting on a different part of the dynamics, restrict $\mathcal{S}$:

- (C1) $\langle i[H_d, H_p] \rangle = 0$ only at energy eigenstates of H_p , which holds when $\{H_p, H_d\}$ generate $su(2^N)$ [18, 19]. This restricts $\mathcal{S}$ via the unitary part of the dynamics, analogously to conditions (1)–(3) of Magann et al. [9].
- (C2) For every excited eigenstate $|E_n\rangle$ ($n > 0$), $\exists k \in \mathcal{K}$ such that $\langle E_n | P_k^\dagger H_p P_k | E_n \rangle \neq E_n$, so that the dissipative term removes excited eigenstates from $\mathcal{S}$ [12]. This holds whenever $[P_k, H_p] \neq 0$ for at least one $k \in \mathcal{K}$.

Both conditions are checkable a priori from the operator structure alone, without knowledge of the eigen-spectrum of H_p , and are verified for all MaxCut and spin-glass instances studied here. Condition (C1) is checked by computing the rank of the Lie algebra generated by $\{iH_p, iH_d\}$ via repeated commutators. Condition (C2) reduces to verifying $[P_k, H_p] \neq 0$ for at least one $k \in \mathcal{K}$, which holds generically whenever the noise model contains Pauli operators outside the stabilizer group of H_p .

We emphasize, however, that a rigorous proof that (C1) and (C2) jointly guarantee $\mathcal{S} = \{|\psi_{\text{gs}}\rangle\langle\psi_{\text{gs}}|\}$ for general mixed-state dynamics remains an open question: the NAFQA trajectory is generically mixed due

to the dissipators, and the full characterization of $\mathcal{S}$ in the open-system setting requires a more detailed analysis connected to the theory of invariant sets of Lindbladian dynamics [12, 20]. This limitation is explicitly acknowledged in both the main text and the SI, and is left for future work.

(ii) The referee notes that the success probability ϕ is well below 1 in Fig. 2(c). We emphasize that this does not indicate a failure of convergence, but rather that the simulation has not yet been carried out to sufficiently large Trotter steps s for convergence to be visible at the sample sizes shown. By contrast, for noisy FQA, it is clear that saturation is well below 1: as demonstrated in Figs. 2(e) and 5(c), both r and ϕ exhibit non-monotonic behavior and decay at large s , as the system drifts toward the maximally mixed state once $A(t)^2 < \Lambda_{\text{noise}}$ (Proposition 1). In NAFQA, Theorem 1 guarantees monotone descent at every layer, so convergence to the ground state is expected at sufficiently large s ; however, according to Eq. (16), the number of trajectories M required to maintain a fixed statistical precision increases exponentially with s .

Changes made. We have added Proposition 1 and Theorem 1 to the main text. We have also introduced and discussed the LaSalle invariant set $\mathcal{S}$ and conditions (C1) and (C2) in the revised manuscript. The checkability of both conditions and their relation to the closed-system FQA convergence conditions of Magann et al. [9] are detailed in Section VI of the SI.

“(ii) Exponential overhead that severely restricts practical scalability. For any realistically sized problem (e.g., $L > 50$), this becomes extremely large. Moreover, the overhead also depends on hardware error rates, placing extremely stringent demands on noise levels. The authors acknowledge this but do not quantify the regime where the method might be useful.”

Response. We thank the referee for raising this important point. We agree that the exponential sampling overhead is a fundamental limitation of NAFQA, and we have addressed it quantitatively in the revised manuscript. We summarize the relevant clarifications below.

(i) The sampling cost of NAFQA is governed by Eq. (16) of the revised manuscript,

$$\mathcal{N}(T) = \exp\left[2 \sum_{k \in \mathcal{K}} \int_0^T |\nu_k(t)| dt\right], \quad (8)$$

which depends solely on the magnitudes of the engineered negative rates $\{\nu_k(t)\}$, and is independent of the intrinsic device error probabilities $\{\lambda_k\}$. In NAFQA, the intrinsic noise channel Λ_s is deliberately retained and incorporated into the Lyapunov feedback law; it is never cancelled or mitigated, so no PEC overhead associated with $\{\lambda_k\}$ is incurred. Therefore the referee’s concern that the overhead places stringent demands on hardware noise levels does not apply: the cost is controlled entirely by the feedback law, not by the hardware error rates. In the limit $\nu_k \rightarrow 0$, one recovers $\mathcal{N}(T) \rightarrow 1$, with no additional cost beyond the base FQA overhead of Ref. [9].

We also note that this exponential cost is not an artifact of our particular implementation but is fundamental in nature. We would like to draw the referee’s attention to the well-known paper by Troyer and Wiese, revealing a direct structural analogy with the fermionic sign problem in quantum Monte Carlo simulations. They established that the fermionic sign problem is NP-hard [8]. This analogy suggests that the exponential overhead for simulating negative Lindblad rates may likewise be a fundamental barrier rather than a practical limitation of current methods. We emphasize that this analogy is structural and does not constitute a formal complexity-theoretic reduction.

(ii) The revised manuscript quantifies the total sampling cost relative to the baseline FQA overhead $\mathcal{N}_s^{\text{FQA}} = \mathcal{O}(mds)$ [9] explicitly as

$$\mathcal{N}_s^{\text{NAFQA}} = \mathcal{O}(mds \cdot |\mathcal{K}| \cdot \mathcal{N}^2(s)), \quad (9)$$

where m is the number of samples per Pauli expectation value, d is the number of Pauli terms in the commutator $i[H_d, H_p]$, and $|\mathcal{K}| \ll 4^N - 1$ under the sparse Pauli-Lindblad model. The factor $|\mathcal{K}|$ enters linearly, while $\mathcal{N}^2$ is exponential in $|\nu_k|s$, regulated by the threshold ν_{max} . The practical regime of utility is therefore characterized by two conditions:

- (i) the sparse noise model holds, i.e., $|\mathcal{K}|$ scales polynomially in N ; and
- (ii) the threshold ν_{max} is chosen sufficiently small to keep $\mathcal{N}^2(s)$ tractable at the circuit depths of interest.

We further emphasize that the net advantage of NAFQA is not a reduction in sampling cost relative to FQA; it does not reduce it. Rather, NAFQA restores the monotone descent guarantee that is otherwise irreversibly lost when FQA is executed on NISQ hardware. Proposition 1 formally establishes that noisy FQA loses the Lyapunov condition $\frac{d}{dt}\langle H_p \rangle \leq 0$ once $A(t)^2 < \Lambda_{\text{noise}}$, causing the system to drift toward the mixed state (further away from the true ground state) regardless of the sampling budget or circuit depth, precisely the non-monotonic behavior shown in Figs. 2(e) and 5. Theorem 1 establishes that NAFQA recovers an unconditional hardware-independent monotone descent at all circuit depths and system sizes, independently of $\{\lambda_k\}$, N , and s . This correctness guarantee, which is structurally unavailable to noisy FQA, is the fundamental motivation for our work.

The exponential QPD cost is the price paid for algorithmic correctness, in the same spirit as the overhead accepted in standard PEC (error mitigation) for unbiased expectation value estimation [6, 10, 11], but serving a qualitatively different purpose: monotone descent assurance rather than noise cancellation.

Changes made. The following additions have been made to the revised manuscript in response to this comment. First, Eq. (16) has been corrected (the original contained a typographical sign error), and a clarification that $\mathcal{N}(T)$ is governed by the magnitudes $|\nu_k(t)|$ of the engineered negative rates has been added to the main text. Second, the total sampling cost $\mathcal{N}_s^{\text{NAFQA}} = \mathcal{O}(mds \cdot |\mathcal{K}| \cdot \mathcal{N}^2(s))$ and a statement of the two conditions characterizing the practical regime of utility have been added to the Discussion section (Eq. (19)). Third, a dedicated paragraph summarizing the practical advantages and limitations of NAFQA has been added to the Discussion section. These additions directly address the referee’s concerns regarding scalability, sampling overhead, and the net practical advantage of the algorithm.

“This approach is closely related to other dissipation-engineered cooling schemes, A brief comparison would help contextualize the advantages and limitations.”

Response. We thank the referee for raising this point. In response to this comment, we have added a dedicated comparison paragraph at the beginning of the Discussion section. The main conceptual distinctions and limitations relative to standard dissipation-engineered cooling are as follows.

(i) In standard dissipative cooling or bath-engineering protocols [3–5], the Lindblad operators and decay rates are fixed and designed to have the target state as the unique steady state of the engineered Lindbladian. The protocol is therefore open-loop: the dissipation is not adjusted as a function of the evolving quantum state. In contrast, NAFQA employs a closed-loop, feedback-adaptive strategy: at each Trotter layer s , the decay rates $\Gamma_k(s+1)$ are updated according to the measured expectation values via the Lyapunov condition $\Gamma_k(s+1) = -(\langle P_k^\dagger H_p P_k \rangle - \langle H_p \rangle)_s$. This layer-dependent, state-dependent tuning is the defining feature of NAFQA and is absent in passive dissipative cooling. Furthermore, NAFQA does not require prior knowledge of the ground state to design the dissipation; the feedback law is constructive and derived entirely from the Hamiltonians H_p , H_d , and the available noise operators P_k .

NAFQA uniquely integrates the noise learning (Pauli-Lindblad model via cycle benchmarking) with the feedback control, so that the physical device noise becomes an input to the Lyapunov law rather than an obstacle to be suppressed. This is qualitatively distinct from cooling schemes that assume full control over the engineered bath.

The decay rates $\Gamma_k(t)$ in NAFQA are time-dependent and can take negative values (non-Markovian regime), which is not a feature of standard cooling protocols. The pseudo-Lindblad quantum trajectory or the quasi-probability decomposition approach is essential for implementing negative rates without ancilla qubits.

(ii) Although NAFQA has a different working principle in comparison to dissipative cooling schemes, it is important to explicitly mention its limitations. First, NAFQA requires noise characterization as a prerequisite: the Pauli-Lindblad error probabilities $\{\lambda_k\}$ must be learned via randomized compiling before the feedback law can be constructed. Second, the QPD implementation of the engineered negative rates incurs an exponential sampling overhead $\mathcal{N}^2(s)$ that grows with circuit depth. Third, the sparse Pauli-Lindblad model assumes that the relevant noise operators $\{P_k\}$ scale polynomially in N ; if this assumption breaks down, the feedback law becomes impractical. These limitations are acknowledged in the revised manuscript and discussed in the context of exponential overhead analysis (Eq. (19)).

Changes made. In the revised manuscript, we have added dedicated paragraphs in the Discussion section addressing the conceptual distinction between NAFQA and dissipative cooling protocols, the practical regime of utility of NAFQA, and its limitations.

“I would like to recommend potential publication, if the authors can adequately respond to the above two questions, and specify the practical advantage of the algorithm.”

Response. We thank the referee for the positive recommendation and for the precise and constructive criticisms. We have addressed both critical concerns in detail in our replies to Comments 1 and 2 above. In addition, we wish to draw the referee’s attention to an important contextual point that further motivates and strengthens the manuscript.

The common assumption in PL models is that decay rates are non-negative, on the grounds of Markovianity. Our work was, to the best of our knowledge, the first to introduce negative decay rates into the Pauli-twirled sparse PL model by combining noise learning with a quasiprobabilistic description and to exploit this regime as a feedback-controlled computational resource. Recent independent works [16, 17] now demonstrate that negative rates in the PL form are not merely a theoretical construct: they arise generically in NISQ devices under Pauli twirling, corroborating the physical regime in which NAFQA operates. This places NAFQA in a much stronger position than described in the original manuscript — rather than introducing negative PL rates as an engineered tool for a specific feedback control purpose, our framework is operating in the physically correct regime for realistic quantum hardware, while most prior work on PL-based error mitigation has constrained itself to the non-negative regime.

Changes made. We have added citations to these works and a brief discussion to the revised Introduction and Discussion sections.

We hope that the referee finds these responses satisfactory and is now in a position to recommend the manuscript for publication.

The authors thank the referees again for their insightful feedback, which helped us improve the conceptual clarity, precision, and reproducibility of the manuscript. We believe that these revisions fully address the referees’ concerns and have substantially strengthened the manuscript.

Sincerely,

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